# Worksheet 1.3 — Equations & Graphs of Polynomial Functions

Grade 12 Advanced Functions (MHF4U) · Unit 1: Polynomial Functions

## The Big Ideas

In factored form $f(x) = a(x-r_1)^{m_1}\cdots$, the zeros are $r_i$ and the multiplicities $m_i$ control how the graph meets the x-axis.

- Zeros: set each factor to 0.
- Odd multiplicity crosses; even multiplicity touches and turns.
- y-intercept $= f(0)$; find $a$ from one extra point.

## Core Concepts

### From zeros to equation

Given the zeros, write the factored form $y = a(x-r_1)(x-r_2)\cdots$; one extra point then fixes the leading coefficient $a$.

### Multiplicity and shape

A factor's multiplicity shapes the graph at its zero: odd crosses the axis, even touches it, and higher multiplicity flattens the curve there.

### Sketching

Combine end behaviour, the zeros with their multiplicities, and the $y$-intercept to sketch the whole polynomial.

# Worked Examples

## Example 1 — Zeros

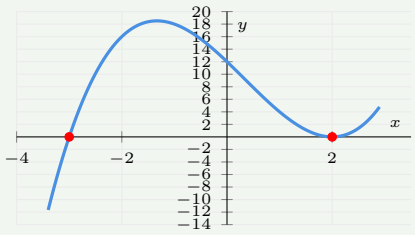
Zeros of $f(x) = (x-3)(x+1)(x-2)$?

***Solution.*** Set each factor to 0: $x = 3, -1, 2$.

## Example 2 — Multiplicity

How does $f(x) = (x+3)(x-2)^2$ meet the x-axis?

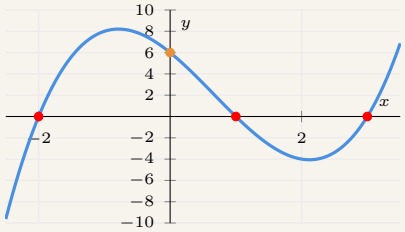
***Solution.*** At $x = -3$ (mult 1) it crosses; at $x = 2$ (mult 2) it touches:



## Example 3 — y-intercept

y-intercept of $f(x) = (x-1)(x+2)(x-3)$?

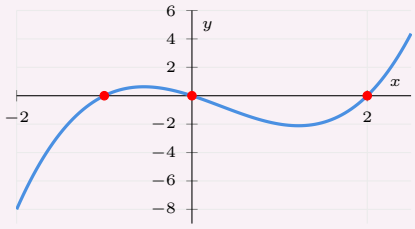
***Solution.*** $f(0) = (-1)(2)(-3) = 6$, so $(0, 6)$:




## Example 4 — Build an equation

Write a cubic with zeros $-1, 0, 2$.

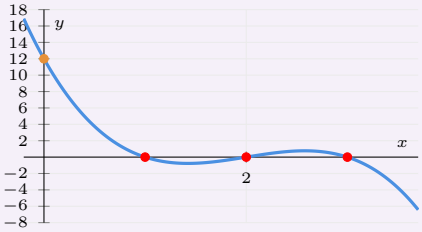
***Solution.*** $f(x) = a\,x(x+1)(x-2)$; e.g. $y = x(x+1)(x-2)$:




## Example 5 — Find a

A cubic has zeros $1, 2, 3$ and passes through $(0, 12)$. Find $a$.

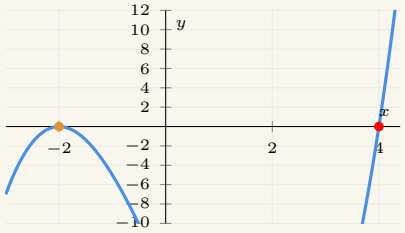
***Solution.*** $12 = a(0-1)(0-2)(0-3) = a(-6) \Rightarrow a = -2$:




## Example 6 — Zeros with multiplicity

Zeros of $f(x) = (x-4)(x+2)^2$ and behaviour?

***Solution.*** $x = 4$ (cross), $x = -2$ (touch):



## Example 7 — Cross or touch

At $x = 3$ in $(x-3)^2(x+1)$?

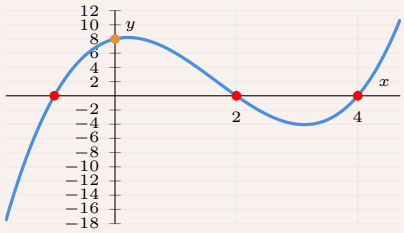
***Solution.*** Even multiplicity (2): touches:




## Example 8 — y-intercept

y-intercept of $(x-2)(x+1)(x-4)$?

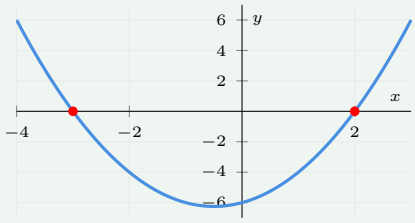
***Solution.*** $(-2)(1)(-4) = 8$:




## Example 9 — Build an equation

Write a polynomial with zeros $-3$ and 2.

***Solution.*** $y = a(x+3)(x-2)$; e.g. $y = (x+3)(x-2)$:



## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

Zeros of $f(x) = (x-4)(x+2)(x-1)$?

### Question 2

At $x = -4$ in $(x+4)^2(x-1)$, cross or touch?

### Question 3

y-intercept of $(x+3)(x-1)(x-5)$?

### Question 4

Write a polynomial with zeros $-3$ and $2$.

### Question 5

A quadratic has zeros $1, 5$ and passes through $(0, 10)$. Find $a$.

## Question 6

Zeros of $f(x) = x(x-3)(x+5)$?

## Question 7

At $x = -2$ in $(x+2)^3$, cross or touch?

## Question 8

y-intercept of $(x+1)(x-2)(x+3)$?

## Question 9

Write a cubic with zeros $0, 2, -2$.

## Question 10

A cubic has zeros $-1, 1, 2$ and passes through $(0, 4)$. Find $a$.

## Question 11

Zeros of $f(x) = (x-5)^2(x+1)$ and their behaviour?

## Question 12

y-intercept of $x(x-1)(x+4)$?

## Question 13 — Challenge

Sketch the sign of $(x+1)(x-2)^2$ across the number line.

# Question Recap & Answer Key

## Question Recap

1. Zeros of $f(x) = (x-4)(x+2)(x-1)$?
2. At $x = -4$ in $(x+4)^2(x-1)$, cross or touch?
3. y-intercept of $(x+3)(x-1)(x-5)$?
4. Write a polynomial with zeros $-3$ and $2$.
5. A quadratic has zeros $1, 5$ and passes through $(0, 10)$. Find $a$.
6. Zeros of $f(x) = x(x-3)(x+5)$?
7. At $x = -2$ in $(x+2)^3$, cross or touch?
8. y-intercept of $(x+1)(x-2)(x+3)$?
9. Write a cubic with zeros $0, 2, -2$.
10. A cubic has zeros $-1, 1, 2$ and passes through $(0, 4)$. Find $a$.
11. Zeros of $f(x) = (x-5)^2(x+1)$ and their behaviour?
12. y-intercept of $x(x-1)(x+4)$?
13. Sketch the sign of $(x+1)(x-2)^2$ across the number line.

## ✓ Answer Key

1. $4, -2, 1$
2. touch
3. 15
4. $y = a(x+3)(x-2)$
5. $a = 2$
6. $0, 3, -5$
7. cross (odd mult 3)
8. $(0, 6)$
9. $y = ax(x-2)(x+2)$
10. $a = -2$
11. 5 (touch), $-1$ (cross)
12. 0
13. neg, then pos for $x > -1$ (touches at 2)